

22. Answer: C

The detected voltage graph represents the superposition of microwaves directly transmitted from transmitter and those that are reflected by the metal sheet.

Maximum interference occurs every time the metal sheet is moved by a distance of 60 mm. This means that the total extra distance covered by the microwaves after the movement of the sheet by 60 mm is equivalent to one wavelength of the microwaves.

Thus $\lambda = 2 \times 60 = 120$ mm

23. Answer: D